
# Technical Whitepaper v7.2

Definitive Edition - May 2026

HumanPulse Foundation, Zug (Switzerland)

# 0. EXECUTIVE ABSTRACT

HumanPulse Protocol is the global human uniqueness layer for the AI era. It is the first decentralized infrastructure for Proof of Unique Humanity that operates without proprietary hardware, without storing biometric data, and without dependence on central authorities. The Proof of Pulse (PoPu) mechanism is built on a simple, defensible Core 3-Layer System - Liveness Detection (rPPG + respiration), Challenge Response, and Zero-Knowledge Proof. All other biometric modalities described in this whitepaper (pupillary reflex, micro-expressions, LiDAR photonics) are optional research modules that are not required for network security. They activate automatically when compatible hardware is present, without changing the user experience. The $HPP token powers a usage-driven, self-sustaining economy. At its core: fixed supply, usage-based fees, validator rewards, and a perpetual DAO Treasury. A dynamic burn adjustment mechanism (governed by the DAO) and anti-whale protections are detailed in the Advanced Tokenomics section. The model aligns with industry benchmarks set by Lido, Bitlayer, and other leading DeFi protocols. HumanPulse is designed for one killer use case first: Sybil resistance for airdrops, DAOs, and token launches - a market that already represents billions in value and urgently needs a privacy-preserving solution. This version 7.1 incorporates a clearly defined MVP scope, refined legal language that presents compliance as a design goal rather than a guarantee, and a product narrative focused on simplicity and adoption.

# 1. PREFACE: THE DIGITAL IDENTITY CRISIS

## 1.1 The Collapse of Proof of Knowledge

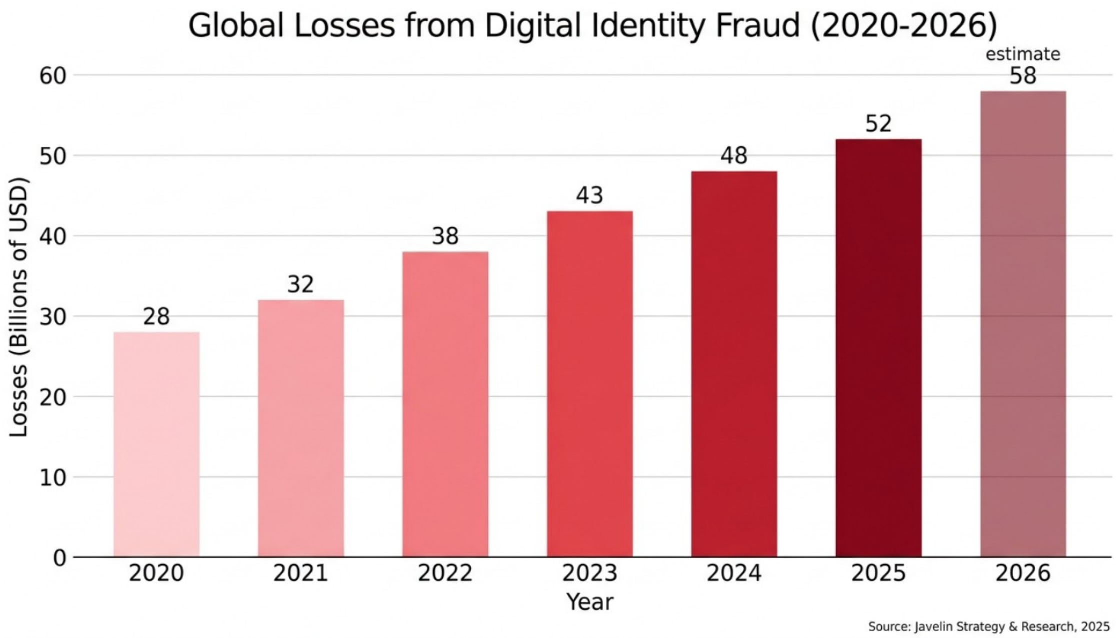


*Figure 1 - Global losses from digital identity fraud (2020-2026).*

By 2026, generative AI has rendered obsolete every authentication system based on secret information. Passwords, digital certificates, scanned documents, and even CAPTCHAs are bypassable with tools accessible to anyone. According to Javelin Strategy & Research (2025), digital identity fraud caused global losses of $52 billion, with an 18% year-over-year increase. The problem is no longer credential security, but the very foundation of identity: information no longer proves humanity.

## 1.2 Biometric Sovereignty against State Surveillance

Systems such as Aadhaar in India (breached in 2023 with over 100 million records exposed) and China's e-ID centralize vast biometric databases, exposing citizens to surveillance and data theft. HumanPulse reverses this paradigm: biometrics becomes a tool for individual self-determination, not control. The paradox is resolved by Zero-Knowledge cryptography: you can prove you are a unique human without revealing anything about your identity.

## 1.3 The Opportunity

Apple, Google, and Microsoft are integrating on-device biometric verification into their closed ecosystems. However, these solutions are tied to commercial interests and are non-interoperable. HumanPulse does not compete with Big Tech: it complements them, offering a neutral, open-source layer governed by a Swiss non-profit foundation. For political voting, CBDCs, and universal basic income, neutrality is an irreplaceable competitive advantage.

2. THE HEART OF THE PROTOCOL: PROOF OF PULSE (PoPu)

## 2.1 Scientific Foundations

Proof of Pulse (PoPu) is the biometric consensus mechanism that distinguishes HumanPulse. It does not replace network consensus (Proof of Stake on Solana), but stands alongside it as an ontological certainty layer: a validator must periodically prove they are alive and unique.

## 2.2 The Core 3-Layer System

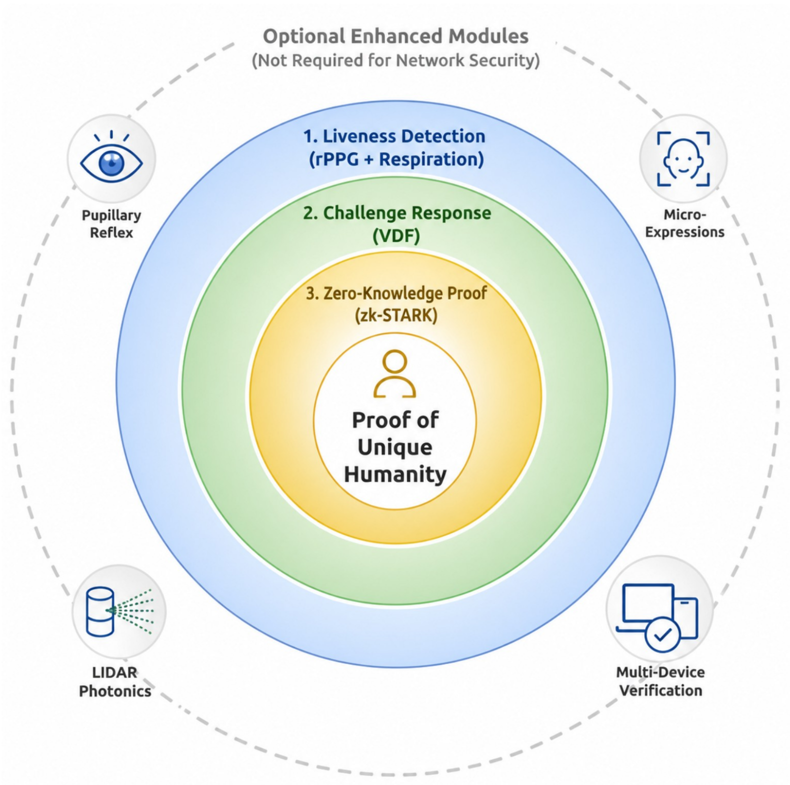


*Figure 2 - Core 3-Layer PoPu architecture. Optional enhanced modules are shown as research modules, not required for network security.*

At its core, HumanPulse is built on three simple, defensible layers that an attacker must defeat simultaneously. Every other biometric modality mentioned in this whitepaper - pupillary reflex, micro-expressions, LiDAR photonics, multi-device verification - is an optional research module, not required for network security. The core system works with any standard webcam.

## Layer 1 - Liveness Detection (rPPG + Respiration)

The user frames their face. The device measures two physiological signals that are impossible for a deepfake to replicate together in real time: heartbeat (rPPG) via micro-chromatic variations of the skin, and respiratory rate via shoulder and chest movement. In May 2025, the Fraunhofer HHI demonstrated that deepfakes can inherit a passive heartbeat signal from the pilot video. HumanPulse solves this by requiring an active response: the heartbeat must modulate in real time in response to a challenge.

## Layer 2 - Challenge Response

The user receives an unpredictable instruction generated on-chain by a Verifiable Delay Function (VDF). This could be as simple as "turn your head slightly" or "read this sentence aloud." The challenge is designed to be effortless - never requiring physical strain or emotional effort - and always offers a passive alternative for accessibility. The physiological response to this challenge is what a deepfake cannot pre-compute.

## Layer 3 - Zero-Knowledge Proof (zk-STARK)

All processing happens inside the device's Trusted Execution Environment (TEE). The raw biometric data is converted into an arithmetic circuit that produces a single, anonymous mathematical proof: "A living, unique human performed challenge X." Only this proof goes on-chain. No biometric data is ever stored, transmitted, or accessible to the protocol. Crucially: no biometric identity is ever created or stored - only a temporary liveness event exists during verification. This distinction is fundamental to both privacy and regulatory compliance.

## 2.3 Optional Research Modules (Not Required for Network Security)

The following modules activate automatically when compatible hardware (LiDAR, high-resolution sensors) is present. They increase anti-spoofing precision but are not required for the core verification flow. They represent research directions for future protocol upgrades, not prerequisites for launch.

| Module | Function | Hardware required |
|---|---|---|
| Pupillary reflex | Measures involuntary pupil dilation in response to light and cognitive load. | Standard camera |
| Micro-expression analysis | Detects involuntary facial muscle activations during verification. | 720p+ camera |
| LiDAR elastic deformation | Measures skin deformation over bone structure when depth hardware is present. | LiDAR / depth sensor |
| Photonic time-of-flight | Verifies real-time photon emission and return timing when LiDAR is active. | LiDAR / depth sensor |
| Multi-device verification | Requires synchronized proof from two heterogeneous devices for high-assurance contexts. | Two compatible devices |

Table 2.3 - Optional modules are explicitly separated from the MVP to preserve execution credibility.

## 2.4 Mathematical Formalization of Security

The core 3-layer system reduces the probability of a successful spoofing attack by orders of magnitude compared to single-modal systems. Each layer introduces independent physiological constraints that a deepfake would need to defeat simultaneously: rPPG with active modulation: a deepfake must generate a coherent, modulated heartbeat signal in real time Respiratory coherence: a deepfake must synchronize chest movement with the heartbeat Challenge response: a deepfake must execute an unpredictable instruction that it cannot pre-compute Combined, these constraints make large-scale automated spoofing computationally and economically prohibitive. For Pulse-Gold (LiDAR + TEE + all enhanced modules active), the security margin increases further. Exact numerical claims are avoided here because real-world attack success rates depend on attacker sophistication, hardware variability, and environmental conditions. What can be stated with confidence is that the multi-signal approach eliminates the single point of failure that makes traditional liveness detection vulnerable.

### 2.5 The User Experience: Invisible Security

From the user's perspective, HumanPulse is frictionless: 1. "Verify your humanity" - Frame your face 2. Follow one simple instruction (e.g., "turn your head slightly" or "read this sentence") 3. Done The entire process takes 10-15 seconds. The user never encounters terms like "STARK," "TEE," "VDF," or "LiDAR." The complexity is entirely hidden under the hood. No biometric identity is ever created or stored - only a temporary liveness event exists during verification.

### 2.6 Attack Resistance

Deepfake with inherited heartbeat: Fails because it cannot modulate dynamically in response to an unpredictable challenge. Hyper-realistic masks: No rPPG signal, no elastic deformation, no pupillary reflex. Physical coercion: The system detects acute stress (altered HRV, mydriasis) and activates duress wallet and differentiated delay without alerting the coercer.

## 3. THE DIFFERENTIATED TRUST PARADIGM

### 3.1 Three Verification Levels

| Level | Minimum hardware | Core system | Primary use cases |
|---|---|---|---|
| Pulse-Bronze | 720p+ webcam or smartphone camera. | Core 3-Layer PoPu: rPPG, respiration, challenge response. | Social networks, gaming, airdrops, DAOs. |
| Pulse-Silver | Depth sensor (Face ID / ToF) when available. | Core 3-Layer plus depth consistency analysis. | DeFi, DAOs, e-commerce, lending. |
| Pulse-Gold | Front LiDAR plus stronger device attestation. | Core 3-Layer plus optional enhanced modules. | UBI, CBDCs, political voting, banking. |

Table 3.1 - Higher verification levels increase assurance without creating hierarchies of personhood.

## 3.2 Human-Kiosks

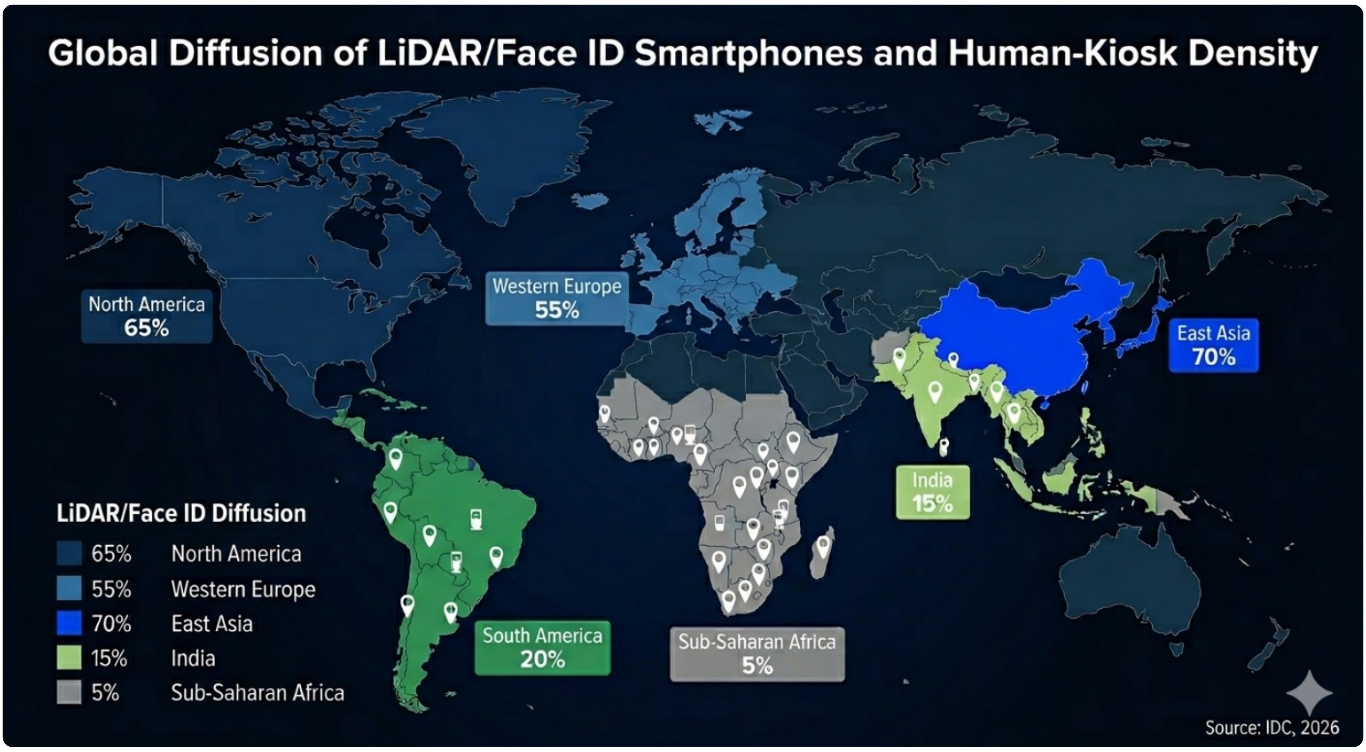


*Figure 3 - Global diffusion of LiDAR/Face ID smartphones and projected Human-Kiosk density.*

A network of physical kiosks based on a self-sustaining social franchising model. Local operators (NGOs, post offices) generate micro-commissions in $HPP for each verification. Robust, low-cost hardware (Intel RealSense, Raspberry Pi, refurbished iPhones) with predictive maintenance.

## 3.3 Offline-First Protocol

The PoPu operates without a constant internet connection. The proof is generated locally and sent when connectivity is available - essential for rural areas and contexts with limited infrastructure.

4. CRYPTOGRAPHIC ARCHITECTURE: zk-STARK

## 4.1 Why STARK and Not SNARK

No Trusted Setup: only collision-resistant hashes (Keccak), no initial secret. Quantum-Resistant: NIST Level 3, resistant to Shor's algorithm. Transparent and publicly verifiable.

## 4.2 STARK Circuit with Hardware Integrity Constraints

The circuit includes constraints that verify: Photonic emission: the LiDAR must prove it is emitting and receiving photons in real time (when LiDAR is active). TEE remote attestation: the TEE produces a manufacturer-signed cryptographic proof that its firmware is intact.

## 4.3 Proof Generation Flow

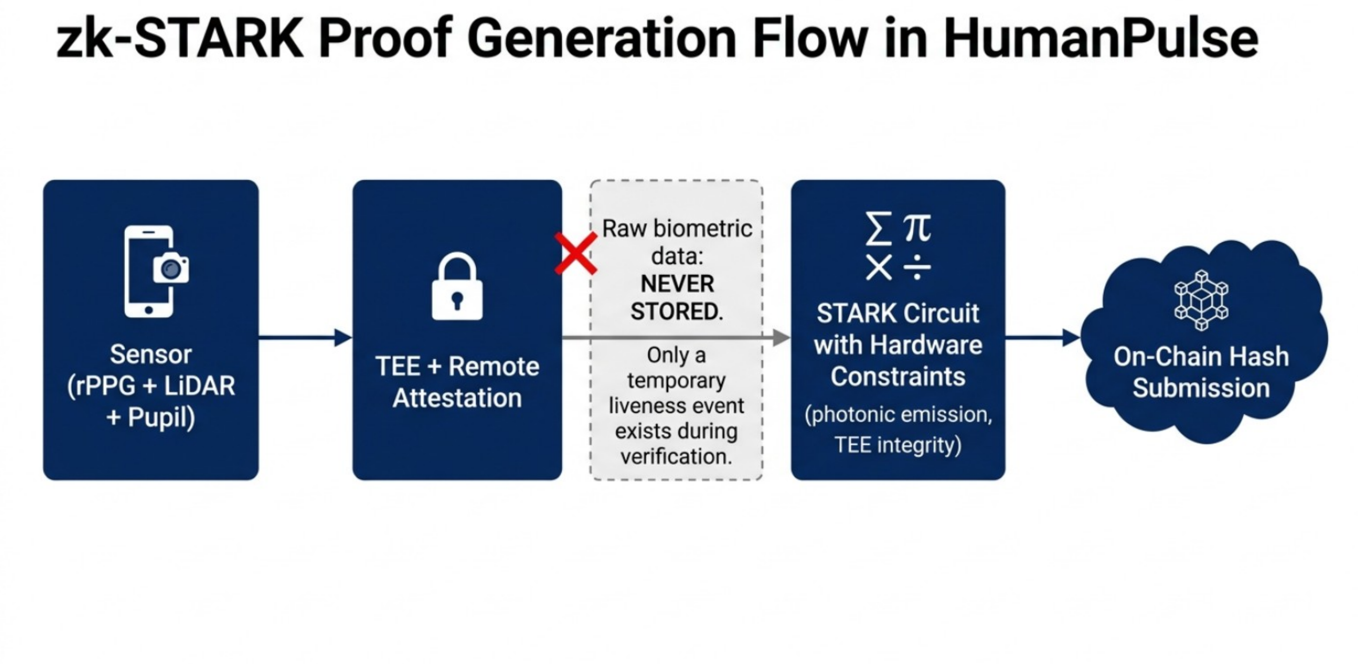


*Figure 4 - zk-STARK proof generation flow with local processing and temporary liveness event.*

Raw data enters the TEE, signals are extracted, the circuit is compiled, and the STARK proof is generated via the FRI protocol. Only the commitment goes on-chain.

## 4.4 STARK Proof Scalability: Recursion and ZK Compression

zk-STARK proofs offer security without a trusted setup, but their size (50-200 kB) may appear as a limitation for on-chain verification on Solana. HumanPulse solves this with a compression and aggregation architecture that drives the marginal cost per verification down to a fraction of a cent.

## A) Recursive Proofs: from N Proofs to 1 Proof

STARK recursion aggregates multiple individual proofs into a single constant-size proof (~50-100 kB). The validator collects N uniqueness proofs, generates a recursive proof attesting "I have verified N valid STARK proofs," and submits it on-chain. This technique, already used by StarkEx (up to 500,000 aggregated transactions) and StarkNet (since version v0.14.2, April 2026), amortizes the verification cost across the entire batch. With N=1,000, the cost per verification drops to 0.1% of the cost of a single verification.

## B) ZK Compression on Solana

Solana features ZK Compression (Light Protocol, Helius), which reduces on-chain storage costs by up to 100x. HumanPulse natively integrates ZK Compression to store uniqueness proofs in compressed form, using specialized indexers (Photon) to query the compressed state.

## C) Asynchronous Aggregation Pipeline

1. The user generates the PoPu proof in the TEE (off-chain). 2. The proof is sent to a validator via the P2P network. 3. The validator aggregates N proofs into a recursive proof (off-chain). 4. The recursive proof is compressed with ZK Compression and verified on-chain on Solana. 5. The verification contract checks the aggregated proof and updates the state.

### D) Cost Efficiency

For a batch of N=1,000 verifications, the amortized on-chain cost per verification is approximately $0.000056. With institutional fees estimated between $0.10 and $1.00 per verification, the operating margin for validators is healthy and the system scales linearly with the number of validators, not with the number of users.

# 5. INFRASTRUCTURE RESILIENCE

## 5.1 Solana Dependency and the Multi-Phase Strategy

Solana offers speed (400 ms block time) and low costs, but has experienced downtime (5 hours in February 2024, other interruptions in 2025). HumanPulse adopts a progressive strategy:

## Phase 1: deferred verification (proofs are queued if the network is down).

## Phase 2 (2027-2028): replication on Ethereum L2 (StarkNet) for redundancy.

## Phase 3 (2029+): dedicated Data Availability layer (Celestia/EigenDA) for uniqueness proofs.

## 5.2 Defense against TEE Attacks (TEE.fail)

The TEE.fail attack (CVE-2025-0520, CVSS 9.4) showed that a DDR5 device costing ~$1,000 can intercept the memory bus and extract secrets from TEEs. Intel and AMD have declared they will not release mitigations.

| Layer | Mechanism | Attacker requirement | Purpose |
|---|---|---|---|
| 1. Remote attestation | TEE produces manufacturer-signed proof of firmware integrity. | Must compromise vendor attestation path. | Confirms software state before proof generation. |
| 2. Anomaly detection | Validator behavior and device history are monitored over time. | Must maintain consistent profile for months. | Detects sudden changes and suspicious clusters. |
| 3. Supply-chain monitoring | Alerts if many identities originate from same chip batch/model. | Must distribute attack geographically. | Reduces scalable batch compromise risk. |
| 4. Multi-device verification | Pulse-Gold can require two devices from different vendors. | Must compromise heterogeneous TEEs simultaneously. | Raises cost for high-assurance use cases. |
| 5. Photonic constraint | LiDAR modules verify real-time photon emission when available. | Must build custom hardware, not only software. | Adds physical barrier beyond emulation. |

Table 5.2 - Hardware defenses are layered; no single TEE is assumed to be perfectly secure.

## 5.3 Countermeasures against Identity Farms

Adaptive geographic cooldown: slows verification from areas with anomalous spikes, without blocking. Reputation decay: inactive identities gradually lose their trust level. On-chain social network coherence: analyzes historical interactions to distinguish real humans from synthetic identities.

# 6. SELECTIVE ATTESTATIONS: PRIVACY-PRESERVING

CREDENTIALS HumanPulse supports zero-knowledge attestations. An authorized entity (e.g., health authority, financial institution, government agency) issues a digitally signed attestation ("over 18," "vaccinated," "accredited investor"). The user stores it locally and generates a STARK proof asserting "I possess a valid attestation meeting criterion X" without revealing the attestation, the issuer, or any personal data. Immediate use cases: KYC for exchanges without submitting documents, age-restricted e-commerce without showing ID, regulated airdrops, DeFi whitelists for accredited investors, political voting with right-to-vote verification without compromising secrecy.

# 7. MVP SCOPE, SDK, AND ADOPTION STRATEGY

## 7.1 MVP Scope: What Ships First

HumanPulse is designed for phased delivery. The initial MVP (Minimum Viable Product) scope is intentionally narrow to maximize execution certainty: MVP Scope (Phase 1 - Q4 2026): Webcam-only (Pulse-Bronze): rPPG + respiration + challenge response + zk-STARK proof Solana on-chain verification (via validators with recursive aggregation)

Sybil resistance widget for DAOs and airdrops Universal zero-knowledge age verification gate Deferred to Phase 2+ (2027 and beyond): Pulse-Silver (depth sensors) and Pulse-Gold (LiDAR + TEE) Enhanced research modules (pupillary reflex, micro-expressions, LiDAR photonics) Multi-chain resilience (Ethereum L2) Human-Kiosks Complete selective attestations (KYC, residency, accredited investor) This phased approach ensures that the core system is deliverable within 12-18 months by a small, focused team.

## 7.2 The Killer Use Case: Sybil Resistance for Airdrops, DAOs, and

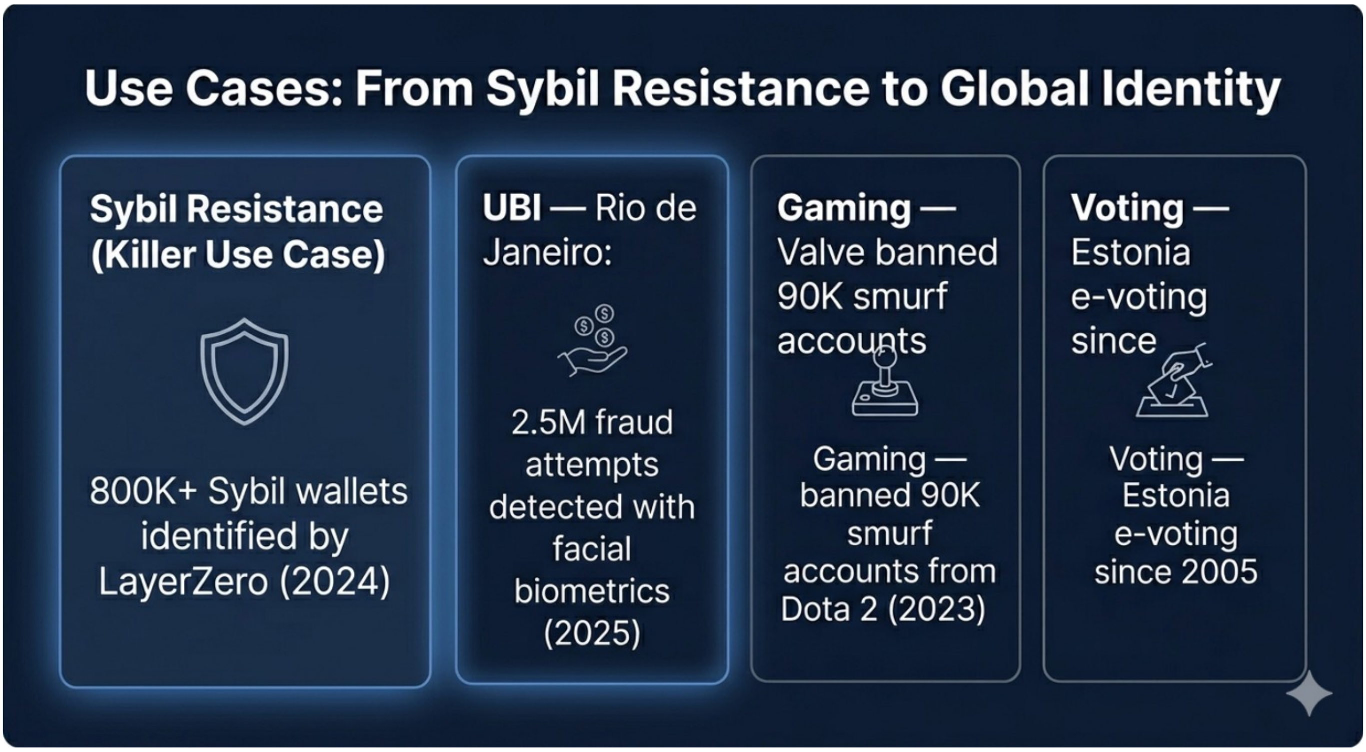


*Figure 7 - Use cases from Sybil resistance to broader human coordination infrastructure.*

Token Launches HumanPulse is designed for one killer use case first: Sybil resistance. Airdrops routinely lose millions to bot farms that spin up thousands of fake wallets. DAOs suffer from governance manipulation by Sybil attackers. Token launches are gamed by automated scripts. With HumanPulse, any DAO, launchpad, or airdrop can integrate a single widget and require every participant to prove they are a unique human before receiving tokens. The verification is privacy-preserving: the DAO never sees the user's identity, only the cryptographic proof of uniqueness. Market validation: In 2024, LayerZero identified over 800,000 Sybil wallets in a single airdrop. The demand for a privacy-preserving Sybil resistance layer is immediate, global, and represents billions in protected value.

## 7.3 SDK Mobile-First

Libraries for iOS, Android, React Native, and Flutter. Core functions: verifyHumanity() and requestAttestation() . Automatic level detection (Bronze/Silver/Gold) based on available hardware.

## 7.4 Widget Web

A single line of HTML. The user clicks, scans a QR code with their smartphone, completes the verification, and the site receives an anonymous confirmation.

## 7.5 The User Experience: Invisible Security

From the user's perspective, HumanPulse is frictionless: 1. "Verify your humanity" - Frame your face 2. Follow one simple instruction (e.g., "turn your head slightly" or "read this sentence") 3. Done The entire process takes 10-15 seconds. No biometric identity is ever created or stored - only a temporary liveness event exists during verification. The user never encounters cryptographic terminology.

## 7.6 Partner Program

The first 100 dApps that integrate HumanPulse receive $HPP incentives from FREI. Institutions (banks, municipalities) have access to dedicated endpoints with guaranteed SLA.## 8. TOKENOMICS OF $HPP

## 8.1 Core Economics (Simple)

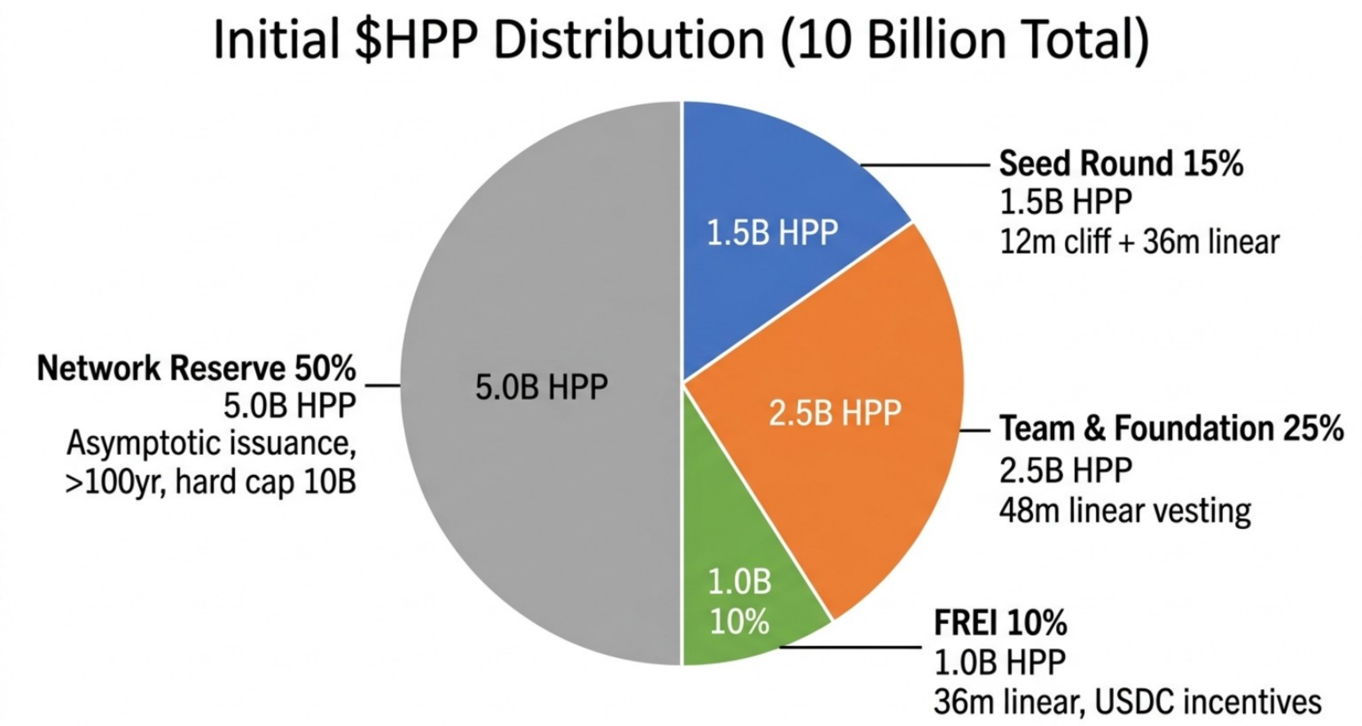


*Figure 6 - Initial HPP distribution with 10 billion total supply.*

The HumanPulse token ($HPP) is a fixed-supply, utility-driven asset. Its value is tied to real-world identity verification demand. Fixed supply: 10 billion $HPP (hard cap, no inflation) Usage-based fees: Every verification generates a fee in $HPP Validator rewards: Validators earn $HPP for processing and aggregating proofs Treasury: A DAO-governed treasury funds ecosystem growth, grants, and partnerships This is the entire model at a glance. Everything else - dynamic burn, anti-whale tax, FREI, asymptotic issuance - is an optimization layer that the DAO can tune over time.

## 8.2 Advanced Tokenomics (DAO-Governed)

The following mechanisms are governed by the DAO and can be adjusted based on network conditions: Fee Distribution (default bootstrap values):

| Allocation | Bootstrap phase | Mature phase | Purpose |
|---|---|---|---|
| Burn | 50% | 20-40% | Permanently removes tokens from circulation and rewards early usage. |
| Validator rewards | 30% | 30-40% | Pays validators for aggregation, verification work, and throughput. |
| DAO treasury | 20% | 30-40% | Funds grants, audits, integrations, ecosystem growth and partnerships. |

Table 8.2 - Fee allocation is simple at the core and adjustable through governance.

Dynamic Burn Adjustment: During the bootstrap phase, a higher burn rate (40-50%) accelerates deflation and rewards early adopters. As the network matures and verification volume grows, the DAO can vote to reduce the burn rate (e.g., to 20-30%) and redirect more resources to the Treasury and validators. This creates a self-balancing economic engine inspired by successful models such as Lido (70/30 fee split with minimum treasury threshold) and Bitlayer (40-30-30 ecosystem allocation). Anti-Whale Tax: Transactions exceeding 1% of total DEX liquidity within 24 hours incur a progressive tax (from 2% to 20%), entirely burned. FREI (Initial Emission Reserve Fund): USDC rewards for early users, differentiated by verification level and region, with progressive decay over 36 months. Maximum estimated cost for 50 million users: approximately $125 million, covered by the seed round. FREI does not mint new $HPP tokens. Asymptotic Issuance: The Network Reserve (5 billion $HPP) is released at an initial rate of 5% per year, decaying by 20% annually, asymptotically approaching zero. The total supply never exceeds 10 billion.

## 8.3 Technical Parameters

| Parameter | Value |
|---|---|
| Name | HumanPulse Protocol |
| Symbol | $HPP |
| Primary blockchain | Solana (Token-2022) |
| Decimals | 9 |
| Initial supply | 10,000,000,000 (10 billion) |
| Maximum supply | 10 billion (hard cap) |
| Mint authority | Foundation Wallet -> DAO (Phase 4) |
| Freeze authority | Revoked |

Table 8.3 - Fixed technical parameters support predictable token behavior.

## 8.4 Initial Distribution

| Allocation | % | Amount | Vesting / release | Purpose |
|---|---|---|---|---|
| Seed Round | 15% | 1.5B HPP | 12-month cliff, 36-month linear | Development, audit, legal |
| FREI | 10% | 1.0B HPP | 36-month linear via smart contract | USDC incentives for users and dApps |
| Team & Foundation | 25% | 2.5B HPP | 48-month linear | Ongoing development and operations |
| Network Reserve | 50% | 5.0B HPP | Asymptotic issuance | Validators, challenges, governance |

Table 8.4 - Distribution separates early development, incentives, operations and reserve.

## 8.5 Demand Drivers

$HPP demand is structurally anchored to: Institutional verification usage (KYC, fintech, Web3 onboarding) Sybil-resistance demand (airdrops, DAOs, token launches) Digital identity infrastructure adoption As integrations grow, demand scales non-linearly.

## 8.6 Supply Dynamics

Supply decreases with every verification No continuous token inflation Early emissions limited to FREI (bootstrap only, in USDC) Over time, circulating supply trends toward scarcity.

## 8.7 Validator-Aligned Economics

Validators are paid through real economic throughput, not inflation. Rewards scale with network usage. No dilution pressure on token holders. Incentives aligned with long-term growth.

## 8.8 Flywheel Effect

HumanPulse is built around a self-reinforcing economic flywheel: More users -> more verifications -> higher fee generation -> more token burn -> reduced supply + growing demand -> value pressure -> more validators join -> better scalability -> back to step 1

### 8.9 Institutional Revenue Bridge

Enterprises pay for verification services in fiat. Revenue is converted into $HPP demand via a dedicated gateway, creating real-world cash flow -> token demand. This is what makes the model VC-relevant, as demonstrated by VeryAI's $10M seed round (Polychain Capital, March 2026) and the broader identity infrastructure sector that attracted over $1.5 billion in venture funding in 2025.

### 8.10 Long-Term Value Thesis

HumanPulse positions $HPP as a deflationary identity infrastructure token, where value is directly derived from verified human presence in the digital world.

## 9. GOVERNANCE: THE POWER OF ONE

### 9.1 Quadratic Voting (QV)

The cost of casting n votes is proportional to n^2, preventing whale dominance.

### 9.2 MACI (Minimal Anti-Collusion Infrastructure)

Users encrypt their vote; an operator produces a ZK proof of the tally without revealing individual votes. Vote keys can be changed anonymously, making vote-buying unverifiable and impractical.

### 9.3 Appealable Blacklist and Social Recovery

Infractions with temporal decay and a decentralized appellate court (21 Pulse-Gold validators). Social recovery with 3-5 trusted contacts and a 72-hour waiting period.

### 9.4 Circuit Breaker and Security Council

An on-chain module monitors anomalous volumes and optionally suspends operations for 24 hours. A Security Council of 9 members (7/9 multisig) can intervene in emergencies for 48 hours, with subsequent DAO ratification. Members are subject to biometric slash.

### 9.5 MACI Fast Track for Urgent Decisions

Reduced voting window of 24 hours and 10% quorum, without compromising anti-collusion security.

### 9.6 Transition to DAO

In Phase 4 (2029-2030), governance fully transitions to the DAO, which also receives the Mint Authority.

## 10. CASE STUDIES

### 10.1 Sybil Resistance: The Killer Use Case

In 2024, LayerZero identified over 800,000 Sybil wallets in a single airdrop. DAOs routinely lose governance integrity to bot farms. Token launches are gamed by automated scripts draining liquidity pools before real users can participate. HumanPulse's widget eliminates this entire attack surface. Every participant proves they are a unique human before receiving tokens. The DAO never sees the user's identity.

### 10.2 Biometric UBI in Brazil

In 2025, Rio de Janeiro used facial biometrics for public transport, detecting 2.5 million fraud attempts. With HumanPulse, governments distribute funds directly to each citizen's uniqueness, eliminating duplication.

### 10.3 Gaming: The Smurf Emergency

Valve banned 90,000 smurf accounts from Dota 2. With HumanPulse, each player has only one verified account; a permanent biometric ban prevents new account creation.

### 10.4 Authentic Social Networks

X (Twitter) removed 1.7 million bots in October 2025. HumanPulse ensures every account corresponds to a unique human, without revealing identity.

### 10.5 Electronic Voting in Estonia

Estonia has used e-voting since 2005, but depends on centralized infrastructure. HumanPulse offers secrecy via zk-STARK, coercion protection, and public verifiability.

## 11. COMPLIANCE AND LEGAL FRAMEWORK

### 11.1 HumanPulse Foundation (Zug, Switzerland)

Non-profit entity under Swiss DLT Act (2021), with $HPP classified as a utility token by FINMA. Switzerland offers political neutrality, clear legislation, and protection from external interference.

### 11.2 Privacy Architecture: In-Device Liveness Computation Engine

HumanPulse is not a "biometric system" in the traditional regulatory sense. It is an in-device liveness computation engine: all biometric signals are processed exclusively within the user's device TEE, exist only transiently in volatile memory (RAM), and are never accessible to the protocol. No biometric identity is ever created or stored - only a temporary liveness event exists during verification. The only output that leaves the device is an anonymous zk-STARK proof - a mathematical attestation of uniqueness - not a biometric identifier. This architecture is designed to minimize exposure under GDPR (Article 9), the EU AI Act, and U.S. state biometric laws including Illinois BIPA, California CCPA/CPRA, and Texas TRAIGA. It also aligns with the accessibility requirements established by the Indian Supreme Court ruling (Pragya Prasun v. Union of India, 2025).

### 11.3 U.S. Multi-State Analysis

Illinois (BIPA): BIPA requires written consent before collecting biometric identifiers. HumanPulse processes all biometric signals exclusively within the user's device and never takes possession of raw biometric data. This architecture is designed to operate outside the scope of BIPA's collection requirements. California (CCPA/CPRA): The CCPA defines biometric information as sensitive personal data. HumanPulse's privacy disclosure explains that no biometric data is ever accessible to the protocol, satisfying CCPA transparency requirements. Texas (TRAIGA): TRAIGA requires consent before capturing a biometric identifier for commercial purposes. HumanPulse does not capture; processing is local and ephemeral. Federal (GENIUS Act / a16z Recommendations): Formal recommendations submitted to the U.S.

Treasury advocate for ZKP-based identity verification, placing HumanPulse at the intersection of advanced regulatory thinking and frontier technology.

### 11.4 Universal Zero-Knowledge Age Verification

HumanPulse implements a universal age gate that ensures no minor can access the biometric verification flow. The user obtains a digitally signed "over 18" attestation from an authorized entity, stores it locally, and generates a zk-STARK proof of age before the PoPu scan is unlocked. This proof reveals nothing about the user's exact age, date of birth, issuing authority, or country of origin. The adoption of 18 years as the universal minimum age threshold satisfies COPPA (USA), GDPR (EU), PIPL (China), DPDPA (India), LGPD (Brazil), and all other international regulations simultaneously.

### 11.5 Lessons from Worldcoin and Legal Shielding

Between 2023 and 2025, authorities in Kenya, Thailand, Indonesia, and Brazil suspended or investigated Worldcoin for privacy violations. In Kenya, the High Court declared biometric data collection illegal, ordered deletion of all data, and established that token payment invalidates consent. HumanPulse is structurally immune to these issues thanks to five pillars: 1. No biometric data collected: The protocol operates without any centralized database. There is nothing to breach, seize, or delete. 2. Consent without consideration for data transfer: FREI distributes a welcome reward in USDC for ecosystem adoption, not in exchange for biometric data. 3. Proactive Data Protection Impact Assessment (DPIA): A complete, public, and verifiable DPIA will be commissioned before public launch. 4. Swiss Foundation with legal shielding: The Swiss jurisdiction isolates the protocol from rulings issued in other jurisdictions. 5. Biometric kill switch: The user may revoke their identity at any time by destroying the local cryptographic key.

# 12. UPDATED COMPETITIVE ANALYSIS

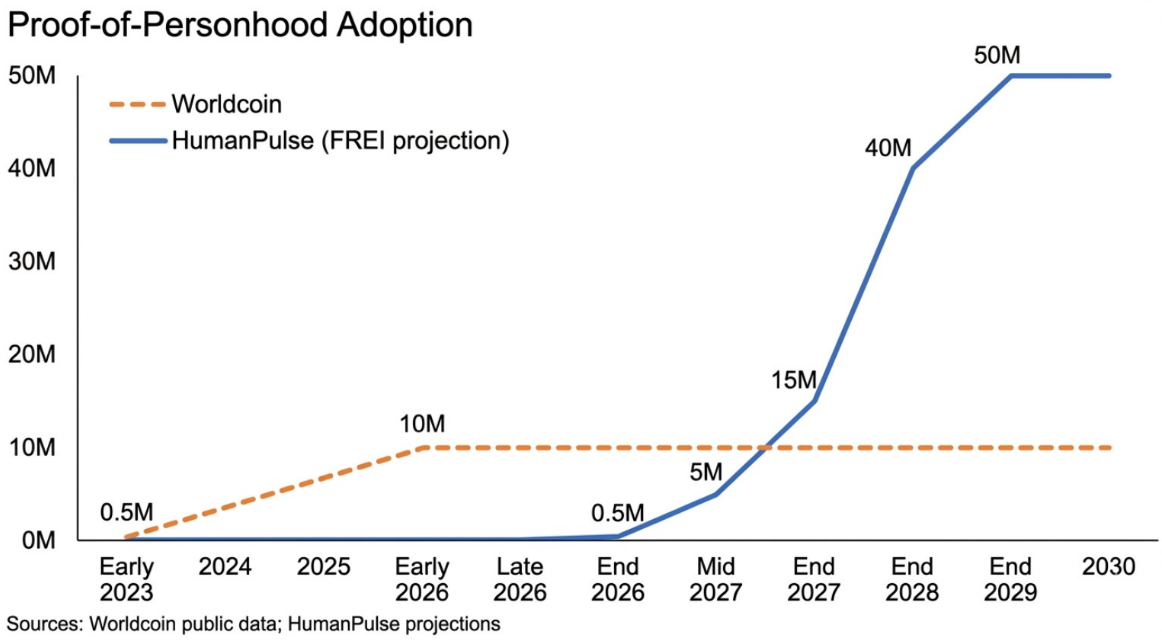


*Figure 5 - Proof-of-personhood adoption projection.*

| Competitor | Technology | Strengths | Weaknesses | HumanPulse advantage |
| --- | --- | --- | --- | --- |
| Worldcoin | Iris scan with Orb, zk-SNARK | Strong uniqueness signal, ZKP narrative. | Proprietary hardware, legal scrutiny, trusted setup concerns. | Smartphone-first, zk-STARK direction, zero biometric storage. |
| Humanity Protocol | Palm scan, Web2 credentials | Less intrusive than iris. | Code opacity and corporate-structure concerns. | Open-source positioning and transparent roadmap. |
| VeryAI | Palm scan via smartphone, ZK proofs | No proprietary hardware, strong backing. | Palm-only modality and weaker multi-signal liveness. | PoPu combines rPPG, respiration and challenge response. |
| Alien | Facial recognition, ZK proofs | Mainnet alpha and public traction. | Invite-only and limited public tokenomics. | Permissionless, transparent tokenomics and Solana-native MVP. |
| Billions | Palm scan via phone, Circom/ZKP | SDK-oriented and document-friendly. | Document dependency and optional liveness. | No documents required; physiological liveness always active. |
| Proof of Humanity | Public video and jury | Decentralized social validation. | Public videos reduce privacy; slower process. | Private proof flow with no on-chain biometric media. |
| Idena | Synchronized CAPTCHA / flip tests | Accessible and gamified. | LLMs can increasingly solve knowledge tests. | Physiological PoPu rather than knowledge-based proof. |

Table 12 - HumanPulse is positioned as smartphone-first proof-of-unique-humanity infrastructure.

## 13. Risk Analysis and Mitigation Matrix

| Risk | Probability | Impact | Mitigation |
|---|---|---|---|
| Deepfake with inherited heartbeat | Medium | High | Active rPPG modulation plus unpredictable challenge and ZK proof flow. |
| TEE.fail / DDR5 attacks | Medium | Critical | Diversified TEE federation, anomaly detection, photonic constraint and multi-device verification. |
| Solana downtime | Medium | Critical | Deferred verification, Ethereum L2 replication and dedicated DA roadmap. |
| Digital divide / exclusion | Medium | Medium | Pulse-Bronze webcam flow, adaptive challenge menu, kiosks and offline-first proof submission. |
| Identity farms | Medium | High | Geographic cooldown, reputation decay and social-coherence anomaly analysis. |
| Validator collusion | Medium | High | Appealable blacklist, biometric slash, anomaly detection and geographic rotation. |
| Big Tech competition | Low | High | Open-source neutrality, Swiss non-profit governance and interoperability. |

Table 13 - Mitigation is layered; the protocol avoids absolute security claims.

| Risk | Probability | Impact | Mitigation |
|---|---|---|---|
| Quantum attack (Shor) | Low (5-10 years) | Catastrophic for unprotected chains | zk-STARK-friendly architecture and migration path to post-quantum signatures (FIPS-204). |
| COPPA / age verification | Low | High (legal) | Universal zero-knowledge age proof with biometric scan gated behind over-18 proof. |
| Regulatory overreach | Medium | High | In-device liveness framing, proactive DPIA, Swiss Foundation and conservative legal language. |

Table 13 (continued) - Regulatory and cryptographic risks remain monitored in the roadmap.

# 14. STRATEGIC ROADMAP 2026-2030

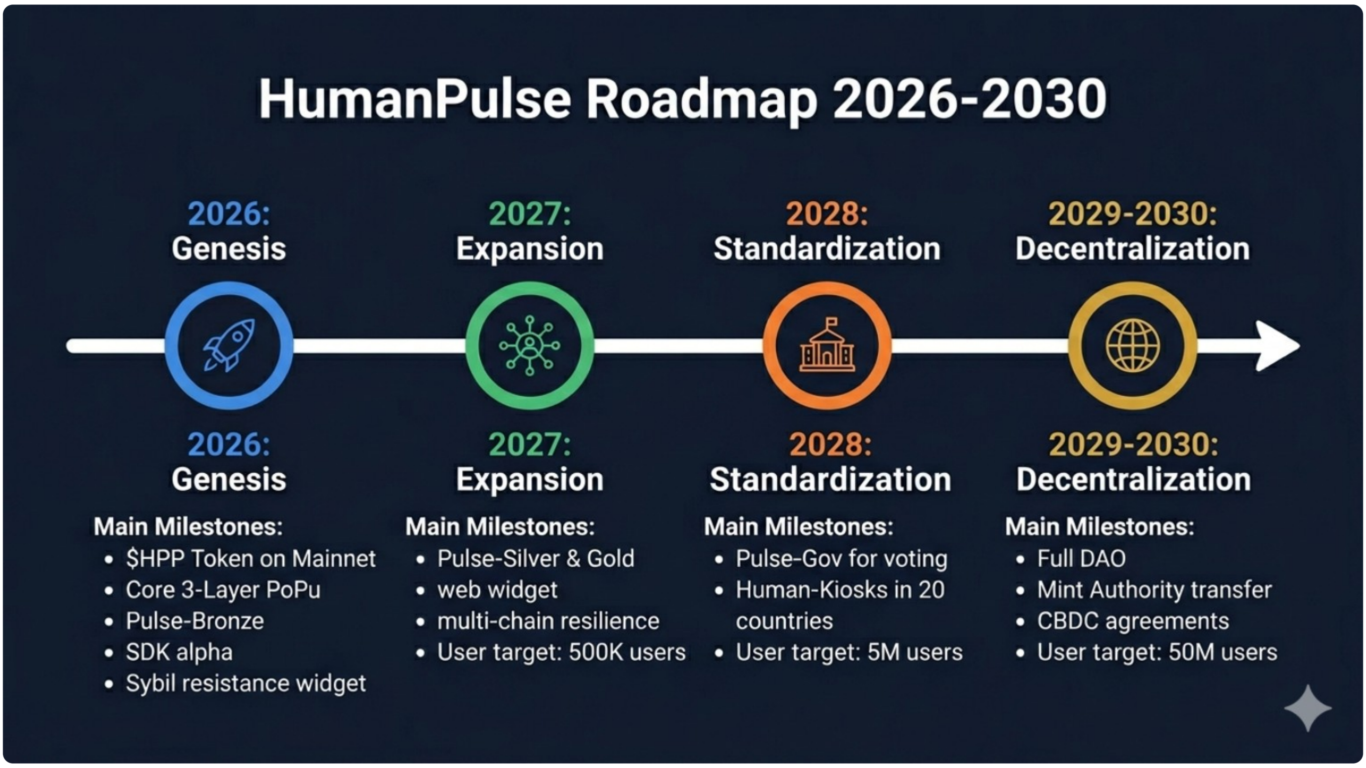


*Figure 8 - HumanPulse roadmap 2026-2030.*

# 14. Strategic Roadmap 2026-2030

The roadmap separates the MVP required for hackathon and early adoption from institutional expansion and long-term decentralization.

| Phase | Period | Milestone | User Target |
|---|---|---|---|
| Phase 1: Genesis | Q4 2026 | $HPP token on Solana mainnet; MVP Core 3-Layer PoPu with webcam; Sybil resistance widget for DAOs and airdrops; universal zero-knowledge age gate; SDK alpha; seed round completed. | 10,000 |
| Phase 2: Expansion | 2027 | Pulse-Silver and Pulse-Gold exploration; optional research modules; multi-chain resilience via Ethereum L2; partner program with 100 incentivized dApps. | 500,000 |
| Phase 3: Standardization | 2028 | Pulse-Gov for voting; Human-Kiosks operational in 20 countries; complete selective attestations for KYC, residency, and accredited investor status. | 5,000,000 |
| Phase 4: Decentralization | 2029-2030 | Full DAO governance; mint authority transfer to DAO; dedicated data availability layer; privacy-preserving CBDC and public-infrastructure partnerships. | 50,000,000 |

*Table 14 - Roadmap milestones separate MVP delivery from institutional expansion and decentralization.*

# 15. Conclusion: The Future of Being

HumanPulse Protocol is designed around a simple thesis: as artificial intelligence makes synthetic participation increasingly cheap, human uniqueness becomes increasingly valuable. It does not try to solve every identity problem immediately; it focuses on proving the presence of a unique living human during digital participation events.

With a simple Core 3-Layer architecture, a clearly defined MVP scope, a usage-driven token model, and a legal framework designed to minimize exposure, HumanPulse is positioned as a privacy-preserving human coordination layer for the AI era.

# Appendix A: Bibliographical References

The references below are maintained as a technical source map for the whitepaper. Some forward-looking or 2026 items should be independently verified before external legal, investment, or regulatory use.

| Reference | Source | Date |
|---|---|---|
| Deepfake study with heartbeat (rPPG) | Fraunhofer HHI, Frontiers | May 2025 |
| TEE.fail attack on DDR5 | CVE-2025-0520, CVSS 9.4 | October 2025 |
| zk-STARK quantum resistance | NIST, FIPS-204 (CRYSTALS-Dilithium) | 2025 |
| zk-STARK verification on Solana mainnet | Cryptology ePrint Archive | October 2025 |
| ZK Compression on Solana | Light Protocol, Helius | 2025 |
| MACI: anti-collusion voting | Aragon, Mechanism Institute | 2025 |
| Global digital identity fraud | Javelin Strategy & Research | 2025 |
| GDPR Article 9 | EU Regulation 2016/679 | 2016 |
| EU AI Act | EU Regulation 2024/1689 | 2024-2025 |
| Indian Supreme Court ruling | Pragya Prasun v. Union of India | April 2025 |
| Worldcoin - Kenya ruling | High Court of Kenya | May 2025 |
| Worldcoin - Kenya data deletion confirmed | ODPC Kenya | January 2026 |
| Solana downtime | Solana Status, Binance | February 2024 |
| Swiss DLT Act | FINMA | 2021 |
| VeryAI - $10M seed round | Polychain Capital | March 2026 |
| Alien - Mainnet alpha launch | Alien Protocol | December 2025 |
| a16z crypto - GENIUS Act ZKP recommendations | a16z crypto | July 2025 |
| FTC - COPPA biometric enforcement policy | Federal Trade Commission | February 2026 |
| Lido - 70/30 fee split with treasury minimum | Lido DAO | 2025 |
| Bitlayer - 40-30-30 ecosystem allocation | Bitlayer | 2025 |
| LayerZero - 800,000 Sybil wallets identified | LayerZero | 2024 |
| Estonian e-voting | e-Estonia | 2005-2025 |
| Valve - 90,000 smurf accounts banned | Forbes, PC Gamer | September 2023 |
| X/Twitter - 1.7 million bots removed | Times of India | October 2025 |
| Rio de Janeiro biometrics | Diario do Rio | 2025 |
| Worldcoin - tokenomics criticism | ZachXBT, CoinMarketCap | 2026 |
| Identity farming and Sybil attack | LayerZero | 2024 |

# Appendix B: Figure List



*End of Technical Whitepaper v7.2 - HumanPulse Protocol ($HPP)*